

Signals & Systems || 2024-2025's Final (1st Semester):

Q1- MCQs:

(30 marks)

Q1) One of the following is Necessary for the Signal $x(t)$ to be Periodic:

- A. The signal repeat itself at equal time interval.
- B. The signal is composed of only one frequency.
- C. The signal amplitude is always constant.
- D. The signal is non-zero for all t .

Q2) for the signal $x(t) = \sin(\frac{2\pi}{3}t)$, Find the fundamental period:

- A. 3
- B. $\frac{3}{2}$
- C. 2π
- D. 6

Q3) for $x[n] = \cos(\frac{\pi}{3}n) + \sin(\frac{\pi}{4}n)$, is the signal periodic, and What is the fundamental period?

- A. Yes, with a period of 12.
- B. Yes, with a period of 8.
- C. Not periodic.
- D. Yes, with a period of 24.

Q4) What is the result of evaluating $\int_{-\infty}^{\infty} e^{-t} \delta'(t) dt$?

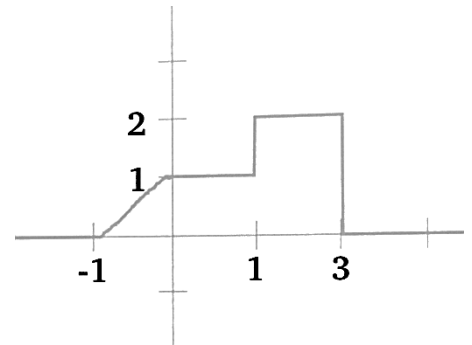
- A. 0
- B. 1
- C. -1
- D. The integral is undefined.

Q5) The result of evaluating $\int_2^3 (t^2 + 2) \delta(t - 1) dt$ is:

- A. 0
- B. 1
- C. 2
- D. 3

Q6) The following signal can be represented as:

- A. $r(t + 1) - r(t) + u(t - 1) - 3u(t - 2)$
- B. $r(t + 1) + u(t) - r(t - 1) - 2u(t - 3)$
- C. $r(t + 1) + r(t) - u(t - 1) - 3u(t - 3)$
- D. $r(t + 1) - r(t) + u(t - 1) - 2u(t - 3)$



Q7) for $\int_{-\infty}^{\infty} (t^2 + \cos(\pi t)) \delta(t - 1) dt$, Which value of t contributes to result?

- A. $t = -1$
- B. $t = 0$
- C. $t = 1$
- D. t contributes over the interval.

Q8) The inverse L.T of $X(s) = \frac{1}{s+1}$, $\text{Re}(s) > -1$ is:

- A. e^t
- B. e^{-t}
- C. te^{-t}
- D. $u(t)$

Q9) The inverse L.T of $X(s) = \frac{s+1}{(s+1)^2+4}$, $\text{Re}(s) > -1$ is:

- A. $e^{-t} \sin(2t)$
- B. $e^{-t} \cos(2t)$
- C. e^{-t}
- D. $e^{-t} \cos(2t) + e^{-t} \sin(2t)$

Q10) The inverse L.T of $X(s) = \frac{2s+4}{s^2+4s-3}$, $\text{Re}(s) > -1$ is:

- A. $2e^{-t} + 4e^{-3t}$
- B. $2e^{-t} - 4e^{-3t}$
- C. $4e^{-t} + 2e^{-3t}$
- D. $e^{-t} + e^{-3t}$

Q11) The $x(t) = e^{-at}u(t) - e^{at}u(-t)$, is:

- A. Even.
- B. Odd.
- C. Neither.
- D. Periodic.

Q12) Which of the following is True for F.S of $x(t) = e^{-at}u(t) - e^{at}u(-t)$:

- A. It contains only sine terms.
- B. It contains only cosine terms.
- C. Both sine and cosine terms.
- D. The F.S does not exist.

Q13) The DC value for the Fourier Series of $x(t) = e^{-at}u(t) - e^{at}u(-t)$, is:

- A. 1
- B. 0
- C. 0.5
- D. All are wrong.

Q14) The inverse of F.T of $X(\omega) = \frac{1}{j\omega+2}$, is:

- A. $e^{-2t}u(t)$
- B. $e^{2t}u(t)$
- C. $e^{-2t}u(-t)$
- D. $e^{2t}u(-t)$

Q15) What is the F.T $x(f)$ of the $x(t) = e^{-2t}u(t)$?

- A. $\frac{1}{2+j2\pi f}$
- B. $\frac{1}{2-2j\pi f}$
- C. $\frac{2}{2+2j\pi f}$
- D. $\frac{1}{2+jf}$

Q2- Find the convolution $y(t) = x(t) * h(t)$ if:

$$x(t) = e^t u(-t), \quad h(t) = e^{-t} u(t)$$

(10 marks)

Q3- Find Laplace transform (using integral) for:

1- $e^t u(-t) - e^{-t} u(t)$

(5 marks)

2- $\text{sgn}(t)$

(5 marks)

Answers:

1- MCQs Answers

1	2	3	4	5	6	7	8	9	10
A	A	D	B	A	D	C	B	B	D
11	12	13	14	15	2 marks each question				
B	D	B	A	A					

2- Find the convolution $y(t) = x(t) * h(t)$ if:

$$x(t) = e^t u(-t), \quad h(t) = e^{-t} u(t)$$

$$x(t) * h(t) = y(t) \Leftrightarrow X(s) \cdot H(s) = Y(s)$$

$$= \mathcal{L}\{e^t u(-t)\} \cdot \mathcal{L}\{e^{-t} u(t)\} = \frac{1}{1-s} \cdot \frac{1}{1+s}$$

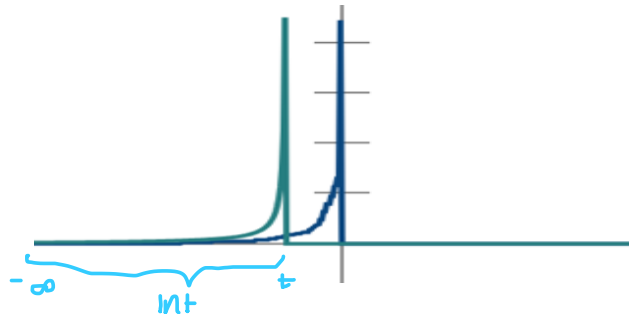
$$= \frac{1}{1-s^2}$$

$$y(t) = \mathcal{L}^{-1}\left\{\frac{1}{1-s^2}\right\} = \frac{1}{2} \mathcal{L}^{-1}\left\{\frac{2}{1-s^2}\right\} = \frac{1}{2} e^{|t|}$$

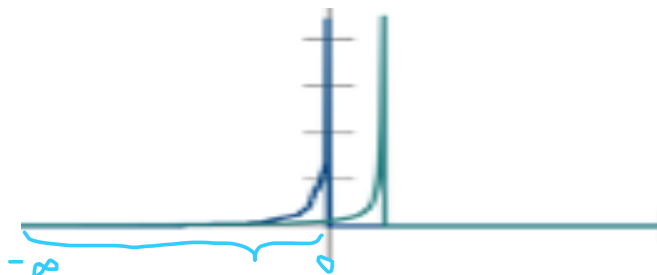
OR By Graphing:

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) \cdot h(t - \tau) d\tau$$

When $t < 0$:



$$\int_{-\infty}^t e^{\tau} \cdot e^{\tau-t} d\tau = \int_{-\infty}^t e^{2\tau-t} d\tau = \frac{e^{-t}}{-2} e^{2\tau} \Big|_{-\infty}^t = \frac{e^t}{2}$$



When $t > 0$:

$$\int_{-\infty}^0 e^{\tau} \cdot e^{\tau-t} d\tau = \int_{-\infty}^0 e^{2\tau-t} d\tau = \frac{e^{-t}}{-2} e^{2\tau} \Big|_{-\infty}^0 = \frac{e^{-t}}{2}$$

Finally:

$$y(t) = \begin{cases} \frac{e^t}{2}, & 0 > t \\ \frac{e^{-t}}{2}, & t > 0 \end{cases} = \frac{e^{|t|}}{2}$$

3- Find Laplace transform (using integral)

for:

$$e^t u(-t) - e^{-t} u(t):$$

$$\mathcal{L}\{e^t u(-t) - e^{-t} u(t)\}$$

$$= \int_{-\infty}^{\infty} e^t u(-t) \cdot e^{-st} dt - \int_{-\infty}^{\infty} e^{-t} u(t) \cdot e^{-st} dt$$

$$= \int_{-\infty}^0 e^t \cdot e^{-st} dt - \int_0^{\infty} e^{-t} \cdot e^{-st} dt$$

$$= \int_{-\infty}^0 e^{t(1-s)} dt - \int_0^{\infty} e^{-t(1+s)} dt$$

$$= \frac{1}{1-s} e^{t(1-s)} \Big|_{-\infty}^0 - \frac{-1}{1+s} e^{-t(1+s)} \Big|_0^{\infty}$$

$$= \frac{1}{1-s} (1 - 0) - \frac{-1}{1+s} (0 - 1) = \frac{1}{1-s} - \frac{1}{1+s}$$

$$= \frac{2s}{1-s^2}$$

$\text{sgn}(t)$:

$$\begin{aligned}\mathcal{L}\{\text{sgn}(t)\} &= \mathcal{L}\{u(t)\} - \mathcal{L}\{u(-t)\} \\&= \int_0^{\infty} 1 \cdot e^{-st} dt - \int_{-\infty}^0 1 \cdot e^{-st} dt \\&= \frac{1}{-s} e^{-st} \Big|_0^{\infty} - \frac{1}{-s} e^{-st} \Big|_{-\infty}^0 \\&= -\frac{1}{s} (0 - 1) - \frac{1}{s} (1 - 0) = \frac{2}{s}\end{aligned}$$

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